ACST1001 Finance Fundamentals – Final Exam Formula Sheet

1. Interest Rates & Inflation (Week 2)

$$r = \frac{1+n}{1+i} - 1$$

- Exact Fisher. n = nominal rate, i = inflation, r = real rate. Use this for precise exam answers.

$$n = (1+r)(1+i) - 1$$

$$r \approx n - i$$

- Approximate Fisher (rough estimate only).

2. Single Cash Flows (Week 3)

Future value (compounding, moving forward in time)

$$FV_n = C(1+r)^n$$

- **Rule 2:** to move a cash flow forward, compound it. r and n must match the compounding period.
- Excel: **=FV(rate, nper, pmt, pv)**, with pmt = 0 and pv entered negative.

Present value (discounting, moving backward in time)

$$PV = \frac{C}{(1+r)^n}$$

- **Rule 3:** to move a cash flow backward, discount it. PV is the worth in dollars today.
- Excel: **=PV(rate, nper, pmt, fv)**, with pmt = 0 and fv entered negative.

3. Multiple Cash Flows, Perpetuities & Annuities (Week 4)

Present value of a cash flow stream

$$PV = C_0 + \frac{C_1}{(1+r)^1} + \frac{C_2}{(1+r)^2} + \cdots + \frac{C_n}{(1+r)^n}$$

- **Rule 1:** cash flows at different times cannot be added until they are moved to the same date. C_0 is not discounted.

Level perpetuity

$$PV = \frac{C}{r}$$

- Gives the value **one period before** the first payment. If the first payment is at Date k, then $PV_0 = (C/r) / (1+r)^{k-1}$.

Growing perpetuity

$$PV = \frac{C_1}{r-g} \quad (r > g)$$

- C_1 is at Date 1, and $C_2 = C_1(1+g)$, $C_3 = C_1(1+g)^2$, and so on. Value is one period before C_1.

Ordinary annuity (first payment at Date 1)

$$PV = C \times \frac{1}{r}\left[1 - \frac{1}{(1+r)^n}\right]$$

$$FV = C \times \frac{1}{r}\left[(1+r)^n - 1\right]$$

- PV is at Date 0. FV is at the date of the last (n-th) payment.
- Annuity due (first payment at Date 0): PV = C_0 + PV of the remaining n−1 period annuity.
- Excel: **=PV(rate, nper, pmt, [fv], [type])** and **=FV(rate, nper, pmt, [pv], [type])**. type = 0 ordinary, 1 due.

4. Solving for Variables, Rates & Loans (Week 5)

Loan / annuity payment

$$C = \frac{PV}{\frac{1}{r}\left[1 - \frac{1}{(1+r)^n}\right]}$$

- PV = amount borrowed. Excel: **=PMT(rate, nper, pv, [fv], [type])**.

Solving for the rate and the number of periods

$$r = \left(\frac{FV}{PV}\right)^{1/n} - 1$$

$$n = \frac{\ln(FV/PV)}{\ln(1+r)}$$

- PV and FV must have opposite signs. Excel: **=RATE(nper, pmt, pv, fv)** and **=NPER(rate, pmt, pv, fv)**.

Rate conversions

$$r_n = (1+r)^n - 1$$

$$EAR = \left(1 + \frac{APR}{m}\right)^m - 1, \qquad r_{period} = \frac{APR}{m}$$

- Excel: **=EFFECT(nominal_rate, npery)**, with nominal_rate = APR and npery = m.

Amortising loans

$$\text{Balance}_k = PV(\text{remaining } n-k \text{ payments})$$

- Balance is taken immediately after payment k.

$$\text{Interest}_k = \text{Opening}_k \times r_{period}$$

$$\text{Principal}_k = C - \text{Interest}_k$$

$$\text{Ending}_k = \text{Opening}_k - \text{Principal}_k$$

5. Bond Valuation (Week 6)

Zero-coupon bond

$$P = \frac{FV}{(1+YTM)^n}$$

- FV is typically $1,000 or $100. YTM and n are per period.

Coupon bond

$$P = CPN \times \frac{1}{y}\left[1 - \frac{1}{(1+y)^n}\right] + \frac{FV}{(1+y)^n}$$

$$CPN = \frac{\text{coupon rate} \times FV}{m}$$

$$y = \frac{APR}{m}, \qquad n = \text{years} \times m$$

- Coupons are an ordinary annuity. The face value is a single payment at Date n.

6. Share Valuation / Dividend Discount Model (Week 7)

General dividend discount model

$$P_0 = \sum_{t=1}^{\infty} \frac{Div_t}{(1+r_E)^t}$$

- r_E = required return on equity.

Constant dividend growth (Gordon growth) model

$$P_0 = \frac{Div_1}{r_E - g} \qquad (r_E > g)$$

$$Div_1 = Div_0(1+g), \qquad P_t = \frac{Div_{t+1}}{r_E - g}$$

7. Risk & Return (Week 8)

$$\text{Total return} = \text{Income return} + \text{Capital return}$$

$$\text{Income return} = \frac{Div_1}{P_0} \quad \text{or} \quad \frac{CPN}{P_0}$$

$$\text{Capital return} = \frac{P_1 - P_0}{P_0}$$

8. Capital Budgeting & Investment Rules (Weeks 9 & 10)

Net present value (NPV)

$$NPV = C_0 + \sum_{t=1}^{n} \frac{C_t}{(1+r)^t}$$

- Outlays are negative. Independent projects: accept if NPV > 0. Mutually exclusive projects: pick the highest positive NPV.

Payback period

$$\text{Payback} = t + \frac{U}{CF}$$

- t = last full year before recovery, U = unrecovered cost at the start of year t+1, CF = cash flow during that year. Accept if within the cutoff. Ignores time value of money and cash flows after the cutoff.

Internal rate of return (IRR)

$$0 = C_0 + \frac{C_1}{1+IRR} + \frac{C_2}{(1+IRR)^2} + \cdots + \frac{C_n}{(1+IRR)^n}$$

- Accept if IRR > cost of capital. If rankings conflict, follow NPV.

Equivalent annual annuity (EAA) / cost (EAC)

$$EAA = \frac{NPV}{\frac{1}{r}\left[1 - \frac{1}{(1+r)^n}\right]}$$

- For projects with different lives that will be replaced. Pick the highest EAA (or lowest EAC).

9. Corporate Equity & IPO Valuation (Week 11)

Venture capital valuation

$$\text{Pre-money} = \text{Pre-round shares} \times \text{Round price}$$

$$\text{Post-money} = \text{Pre-money} + \text{Capital raised}$$

$$\text{Ownership} = \frac{\text{Shares owned}}{\text{Total post-round shares}} \times 100\%$$

IPO valuation & cost

$$\text{Equity value} = \text{Net profit} \times \text{P/E}$$

$$\text{IPO price} = \frac{\text{Equity value}}{\text{Post-IPO shares}}$$

$$\text{Underpricing} = \frac{P_{day1} - P_{offer}}{P_{offer}}$$

$$\text{Fee} = (\text{Shares sold} \times P_{offer}) \times \text{Spread}$$

- Underwriting fee. Spread is the underwriting spread (%).